

JIAQI WU

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Education

Tsinghua University, Department of Computer Science and Technology

Sep. 2022 – Present

Senior Undergraduate, Computer Science and Technology

GPA: 3.98/4.0; 104 credits with grades of A or above

Ranking: 3/201

Research Interests

Computer Graphics, Computer Vision, Rendering Techniques

Publications

- Gradient Domain Reconstruction for Monte Carlo PDE Solvers
Jiaqi Wu, Xuejun Hu, Shuang Zhao, Kun Xu
ACM Transactions on Graphics (SIGGRAPH 2026)
- Generalized Spherical Harmonics Products using Spherical Grids
Di An, **Jiaqi Wu**, Bowen Xu, Lingqi Yan, Kun Xu
ACM Transactions on Graphics (SIGGRAPH 2026)
- Adding Regional Control for Continuous Remeshing via Attention Flows
Jiaqi Wu, Kun Xu
SIGGRAPH 2025 Poster
- SkinRig: Skinning-Prior-Guided Skeleton Binding for Human Meshes
Jiaqi Wu*, Wanxi Dong*, Yixin Jin, Kun Xu
Under review

Research Experience

The 5th CGAI Challenge

Team Leader and First Author

Beijing, China

May 2025 – Present

► **Human Skeleton Generation Track**

- Led the team and ranked first on both the public and private leaderboards; received the First Prize and was invited to submit a CVMJ Short Communication. The related work is being prepared for submission to a top international journal or conference.
- Extracted features using a PCT network and trained a decoder to regress skinning weights for each vertex, producing stable weight fields.
- First introduced predicted skinning weights as conditional input to the joint-position prediction head, improving robustness and generalization under limited training data.

Beijing Undergraduate Research Program

Project Leader

Beijing, China

May 2025 – Present

► **Efficient PDE Solving Framework Based on Grid-free Monte Carlo and Sample Reuse**

- Initiated the project and received the highest level of funding; developed an efficient PDE solving framework based on grid-free Monte Carlo methods and sample reuse.
- Optimized algorithms to improve the efficiency of partial differential equation solvers for complex physical simulation scenarios.

Tencent Spark Program – Supernova Project

First Author

China

Apr. 2024 – Jun. 2025

► **Adding Regional Control for Continuous Remeshing via Attention Flows**

- Published the work as a SIGGRAPH Poster as the first author; filed one related patent application.
- Developed an attention-flow-based method for regional control in continuous remeshing; **won the Third Prize in the Information Technology Track of the 43rd Tsinghua Challenge Cup.**

Huawei HarmonyOS Pipeline AI Rendering Project

Developer

► Accelerated Rendering via AI-based Indirect Illumination Prediction

• Participated in dataset processing, on-device deployment, and methodological improvement; used AI models to predict indirect illumination for accelerated rendering.

• Optimized the rendering pipeline to improve real-time rendering performance.

China

Aug. 2024 – Jun. 2025

Internship Experience

Huawei

Research Intern

• Studied sparse volumetric voxel rasterization and compared it with 3D Gaussian Splatting.

Beijing, China

Jun. 2025 – Aug. 2025

Tencent

Research Intern, Durer Group

• Developed an attention-flow-based method for regional control in continuous remeshing.

Beijing, China

Jun. 2024 – Oct. 2024

Scholarships and Honors

National Scholarship	Sep. 2025
National Scholarship	Sep. 2024
Tang Zhongying Moral Education Scholarship	May 2024
Tsinghua University Comprehensive Scholarship: Friends of Tsinghua – PetroChina Scholarship	Sep. 2023
Tang Zhongying Moral Education Scholarship	Jun. 2023

Technical Skills

Programming and Tools: Python, C++, PyTorch
Research Areas: Computer Graphics, Computer Vision, Rendering Techniques
Languages: English (CET-4: 636; proficient in reading English academic papers), Chinese (Native)
Other Skills: Teamwork, Project Management, Sports (Long-distance Running, Table Tennis, Swimming)